

Geophysics 3A — Week 5 Workshop

Isostasy: Ocean–Continent Structure & Isostatic Gravity

May 27, 2026

Workshop Overview

This workshop develops a quantitative understanding of isostatic equilibrium and its gravity expression by combining column-based reasoning, ocean–continent structure, and slab gravity approximations. Airy and Pratt isostasy are treated as end-member models, with emphasis on why real Earth structure requires a mixed or layered approach involving the lithospheric mantle.

Duration: 3 hours

Format: Small-group whiteboard work, guided derivations, synthesis discussion

Resources: Provided activity PDFs (referred to during the workshop)

Learning Outcomes

By the end of this workshop, you should be able to:

- Explain isostatic equilibrium as equal lithostatic pressure (or mass per unit area) at a compensation depth.
- Derive and interpret Airy and Pratt isostasy as conceptual end members.
- Explain why crust-only isostatic models fail for ocean–continent systems.
- Derive the Airy isostatic gravity correction using slab approximations.
- Explain why gravity anomalies over high topography may be small.

0–10 min Warm-Up: Defining Isostasy

Individually, write a one-sentence definition of isostatic equilibrium. These definitions are compared in pairs and then within your group, with the goal of converging on a precise statement.

Target idea:

Isostatic equilibrium requires equal lithostatic pressure (or mass per unit area) at a compensation depth beneath adjacent columns.

Groups assign roles (facilitator, skeptic, recorder, assumption checker).

10–30 min Mini-Lecture: Airy, Pratt, and Mixed Isostasy

A short instructor-led lecture reviews:

- Airy isostasy: variable crustal thickness, uniform density
- Pratt isostasy: constant thickness, lateral density variations
- Why neither end member fully describes the Earth
- The role of lithospheric mantle and partial compensation

This provides the conceptual framework for the two analytical activities that follow.

30–70 min Activity 1: Oceanic vs Continental Columns

(Based on the “Oceanic and Continental Crust” activity PDF)

Purpose

To test whether crust-only Airy isostasy can account for observed ocean–continent elevation differences, and to explore how including the lithospheric mantle resolves inconsistencies.

Crust-Only Airy Isostasy

Each group balances an oceanic and a continental column under the assumption that compensation occurs solely through crustal thickness. You will:

- Enforce equal mass per unit area above a compensation depth.
- Solve for the oceanic crust density required for equilibrium.
- Compare your result with realistic oceanic crust densities.

You should find that the required density is unrealistic, indicating a missing component in the model.

Two-Layer Lithosphere Extension

The model is then extended to include a lithospheric mantle beneath both oceanic and continental crust. You will:

- Rebalance the columns including mantle lithosphere thickness differences.
- Estimate the density contrast between oceanic and continental lithospheric mantle.
- Relate this contrast to mineralogical differences (e.g. garnet content).

70–80 min Whole-Class Synthesis

A brief instructor-led discussion synthesises results from Activity 1, focusing on:

- Why crust-only Airy isostasy fails
- Why the lithospheric mantle must be included
- Where Pratt-like behaviour may arise

80–120 min Activity 2: Isostatic Gravity Correction

(Based on the “Isostatic Gravity Correction” activity PDF)

Purpose

To derive the Airy isostatic gravity correction and explain why large topography does not necessarily produce large gravity anomalies.

Airy Mass Balance

You will show that, under Airy isostasy,

$$\rho_c h = (\rho_m - \rho_c) r,$$

and derive an expression for the crustal root thickness $r(x)$.

Infinite Slab Gravity

Using the slab approximation,

$$\Delta g = 2\pi G \rho t,$$

you will compute the gravitational effects of surface topography and of the compensating crustal root.

Cancellation

By substitution, you will demonstrate that under perfect Airy compensation, the gravity effects of topography and root cancel exactly, and discuss what assumptions are required for this result.

120–145 min Numerical Demonstration

The instructor presents a live Python demonstration illustrating:

- Airy compensation with varying compensation fractions
- Pratt-style lateral density variations
- Resulting gravity anomalies and residuals

Interpretation of results is emphasised rather than coding details.

145–170 min Group Synthesis

Each group prepares a brief summary addressing:

- One reason crust-only Airy isostasy fails.
- One way lithospheric mantle structure resolves this failure.
- One equation linking topography, density, and gravity.
- One observation that could reduce non-uniqueness in isostatic interpretation.

170–180 min Exit Questions

Individually, respond to the following:

1. Identify one unrealistic assumption in either Airy or Pratt isostasy and explain why it fails.
2. Explain how geological or geophysical constraints (e.g. seismology or petrology) help resolve ambiguity in isostatic models.

End of Week 5 Workshop